Percentile Ranks

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Introduction

A percentile rank answers "what fraction of observations fall at or below this value?" It converts an absolute measurement to a relative position in the sample distribution. In paediatrics, growth charts give a child's weight percentile; in cognitive testing, an IQ score comes with a percentile rank; in bioinformatics, a gene's expression is often reported as a percentile across a reference panel.

Prerequisites

The reader should know what a quantile is and be comfortable with sorted vectors in R.

Theory

For a sample $x_1, \ldots, x_n$, the percentile rank of a value y is

$$\mathrm{PR}(y) = 100 \cdot \frac{\#\{x_i \le y\}}{n}.$$

Several refinements exist: the "mid-rank" form counts half of any ties; the "plotting position" form uses $(i-a)/(n+1-2a)$ for various a (e.g., Blom's $a = 3/8$).

For a single value, computing the percentile rank is the inverse of computing a quantile: if the 75th percentile is $q_{0.75}$, then $\mathrm{PR}(q_{0.75}) \approx 75$.

Assumptions

Purely descriptive; assumes only that the reference distribution is appropriate to the value being ranked.

R Implementation

```r
library(dplyr)

set.seed(2026)
growth_chart <- rnorm(1000, mean = 15.0, sd = 2.2)   # reference weights (kg)
patient      <- 17.8                                  # a patient's weight

pr_patient <- 100 * mean(growth_chart <= patient)
pr_patient

# Percentile ranks for all observations in a cohort
cohort <- data.frame(weight = rnorm(30, 15.5, 2.0))
cohort$rank    <- rank(cohort$weight)
cohort$pr      <- 100 * cohort$rank / nrow(cohort)
cohort$pr_mid  <- 100 * (rank(cohort$weight) - 0.5) / nrow(cohort)
head(cohort)
```

```r
# ECDF equivalent
ec <- ecdf(growth_chart)
100 * ec(patient)
```

Output & Results

```
[1] 89.4     # 17.8 kg is at the 89th percentile of the reference

  weight rank    pr pr_mid
1  13.29    1   3.3   1.67
2  14.15    6  20.0  18.33
3  15.01   11  36.7  35.00
```

Interpretation

Report percentile ranks in growth, development, and norming contexts. In a clinical note: "body weight 17.8 kg (89th percentile for age/sex)". The percentile tells the reader how unusual the value is relative to the reference.

Practical Tips

- Percentile ranks depend on the reference sample; always state which reference was used (national norms, internal cohort, etc.).
- Ties are handled by `rank()` via averaging, minimum, or maximum; specify `ties.method` when the convention matters.
- For small reference samples, percentile ranks are unstable; use smoothing (LMS method) for growth chart applications.
- A percentile rank of exactly 100 is usually not reported; "above the 99th percentile" is the conventional phrasing for the sample maximum.
- Do not confuse percentile rank (position in a distribution) with percentile (value at a specified position).

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Descriptive statistics and Table 1